

1(a)

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. reg educ sibs brthord meduc married exper tenure
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Source	SS	df	MS	Number of obs	=	784
Model	1207.07633	6	201.179389	F(6, 777)	=	60.37
Residual	2589.16346	777	3.33225671	Prob > F	=	0.0000
				R-squared	=	0.3180
				Adj R-squared	=	0.3127
Total	3796.2398	783	4.84832669	Root MSE	=	1.8254

educ	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
sibs	-.1214585	.0366418	-3.31	0.001	-.1933872	-.0495299
brthord	-.0348487	.052545	-0.66	0.507	-.1379957	.0682983
meduc	.1870618	.024385	7.67	0.000	.1391936	.23493
married	-.1678703	.2230105	-0.75	0.452	-.6056446	.2699041
exper	-.2234794	.0161823	-13.81	0.000	-.2552456	-.1917132
tenure	.0361026	.013444	2.69	0.007	.0097117	.0624936
_cons	14.48134	.4263677	33.96	0.000	13.64437	15.31831

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. test sibs brthord meduc
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( 1) sibs = 0
( 2) brthord = 0
( 3) meduc = 0
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F( 3, 777) = 35.59
Prob > F = 0.0000
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(b)

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. ivreg lwage (educ=sibs brthord meduc) married exper tenure
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Instrumental variables (2SLS) regression

Source	SS	df	MS	Number of obs	=	784
Model	7.5798416	4	1.8949604	F(4, 779)	=	22.26
Residual	124.631146	779	.159988634	Prob > F	=	0.0000
				R-squared	=	0.0573
				Adj R-squared	=	0.0525
Total	132.210988	783	.168851836	Root MSE	=	.39999

lwage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
educ	.1536492	.0212051	7.25	0.000	.1120232	.1952752
married	.2054959	.0489864	4.19	0.000	.1093349	.301657
exper	.0353705	.0063778	5.55	0.000	.0228508	.0478902
tenure	.0065216	.0030897	2.11	0.035	.0004564	.0125868
_cons	4.07966	.3511419	11.62	0.000	3.390364	4.768957

Instrumented: educ

Instruments: married exper tenure sibs brthord meduc

(c)

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. predict uhat_iv, r
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. reg uhat_iv sibs brthord meduc married exper tenure
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Source	SS	df	MS	Number of obs	=	784
Model	.09546904	6	.015911507	F(6, 777)	=	0.10
Residual	124.535678	777	.160277577	Prob > F	=	0.9965
				R-squared	=	0.0008
				Adj R-squared	=	-0.0070
Total	124.631147	783	.159171324	Root MSE	=	.40035

uhat_iv	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
sibs	.0049992	.0080361	0.62	0.534	-.0107758	.0207742
brthord	-.0010237	.0115239	-0.09	0.929	-.0236454	.0215979
meduc	.0025283	.005348	0.47	0.637	-.0079699	.0130265
married	-.0003569	.0489094	-0.01	0.994	-.0963672	.0956533
exper	.0001807	.003549	0.05	0.959	-.0067861	.0071474
tenure	.0000795	.0029485	0.03	0.978	-.0057084	.0058674
_cons	-.0413568	.0935086	-0.44	0.658	-.2249162	.1422026

```
. test sibs brthord meduc
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```
( 1) sibs = 0
( 2) brthord = 0
( 3) meduc = 0
```

```
F( 3, 777) = 0.20
Prob > F = 0.8974
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$J = mF = 3 \times 0.2 = 0.6 < x^2(2) = 5.99$. We do not reject the null hypothesis.

(d)

```
. ivreg lwage (educ=sibs brthord meduc) married exper tenure
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Instrumental variables (2SLS) regression

Source	SS	df	MS	Number of obs	=	784
Model	7.5798416	4	1.8949604	F(4, 779)	=	22.26
Residual	124.631146	779	.159988634	Prob > F	=	0.0000
				R-squared	=	0.0573
				Adj R-squared	=	0.0525
Total	132.210988	783	.168851836	Root MSE	=	.39999

lwage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
educ	.1536492	.0212051	7.25	0.000	.1120232	.1952752
married	.2054959	.0489864	4.19	0.000	.1093349	.301657
exper	.0353705	.0063778	5.55	0.000	.0228508	.0478902
tenure	.0065216	.0030897	2.11	0.035	.0004564	.0125868
_cons	4.07966	.3511419	11.62	0.000	3.390364	4.768957

Instrumented: educ

Instruments: married exper tenure sibs brthord meduc

```
. estimate store my2SLS
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```
. reg lwage educ married exper tenure
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Source	SS	df	MS	Number of obs	=	935
Model	29.1888221	4	7.29720551	F(4, 930)	=	49.73
Residual	136.467461	930	.146739206	Prob > F	=	0.0000
				R-squared	=	0.1762
				Adj R-squared	=	0.1727
Total	165.656283	934	.177362188	Root MSE	=	.38307

lwage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
educ	.0753568	.0064349	11.71	0.000	.0627282	.0879854
married	.1991709	.0408196	4.88	0.000	.1190617	.2792802
exper	.0141191	.0033383	4.23	0.000	.0075677	.0206705
tenure	.0127554	.0025592	4.98	0.000	.0077329	.017778
_cons	5.330651	.1143785	46.61	0.000	5.106181	5.555121

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. estimate store myOLS
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. hausman my2SLS myOLS
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	Coefficients		(b-B) Difference	sqrt(diag(V_b-V_B)) S.E.
	(b) my2SLS	(B) myOLS		
educ	.1536492	.0753568	.0782924	.0202052
married	.2054959	.1991709	.006325	.0270819
exper	.0353705	.0141191	.0212514	.0054344
tenure	.0065216	.0127554	-.0062338	.0017311

b = consistent under Ho and Ha; obtained from ivreg
 B = inconsistent under Ha, efficient under Ho; obtained from regress

Test: Ho: difference in coefficients not systematic

chi2(4) = (b-B)'[(V_b-V_B)^(-1)](b-B)
 = 18.43
 Prob>chi2 = 0.0010

Reject the null. Choose IV.

2 (a)

$$\begin{aligned}
 \hat{\beta}_{1,OLS} &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})} = \beta_1 + \frac{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})u_i}{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})x_i} \approx \beta_1 + \frac{Cov(x, u)}{Var(x)} \\
 &= \beta_1 + \frac{\frac{Cov(x, u)}{\sqrt{Var(x)} \times \sqrt{Var(u)}} \times \frac{\sqrt{Var(x)} \times \sqrt{Var(u)}}{1}}{\frac{Var(x)}{\sqrt{Var(x)} \times \sqrt{Var(x)}} \times \frac{\sqrt{Var(x)} \times \sqrt{Var(x)}}{1}} \\
 &= \beta_1 + Corr(x, u) \times \frac{\sqrt{Var(u)}}{\sqrt{Var(x)}} = \beta_1 + Corr(x, u) \times \frac{\sigma_u}{\sigma_x}
 \end{aligned}$$

$$\begin{aligned}
(b) \hat{\beta}_{1,IV} &= \frac{\sum_{i=1}^n (z_i - \bar{z}) y_i}{\sum_{i=1}^n (z_i - \bar{z}) x_i} = \beta_1 + \frac{\frac{1}{n-1} \sum_{i=1}^n (z_i - \bar{z}) u_i}{\frac{1}{n-1} \sum_{i=1}^n (z_i - \bar{z}) x_i} \\
&\approx \beta_1 + \frac{Cov(z, u)}{Cov(z, x)} = \beta_1 + \frac{\frac{Cov(z, u)}{\sqrt{Var(z)} \times \sqrt{Var(u)}} \times \frac{\sqrt{Var(z)} \times \sqrt{Var(u)}}{1}}{\frac{Cov(z, x)}{\sqrt{Var(z)} \times \sqrt{Var(x)}} \times \frac{\sqrt{Var(z)} \times \sqrt{Var(x)}}{1}} \\
&= \beta_1 + \frac{Corr(z, u)}{Corr(z, x)} \times \frac{\sqrt{Var(u)}}{\sqrt{Var(x)}} = \beta_1 + \frac{Corr(z, u)}{Corr(z, x)} \times \frac{\sigma_u}{\sigma_x}
\end{aligned}$$

(c)

Clearly, if x is an endogenous variable, then we should use IV provided that we have instrument exogeneity and instrument relevance. But if the instrument exogeneity does not hold, we will have a large bias when z is a weak instrument.

More specifically, it could be the case $\frac{Corr(z, u)}{Corr(z, x)} > Corr(x, u)$. The situation becomes even worse, when Z and X have a very weak relationship, say for example $Corr(z, x) = 0.05$, In this case, using IVs is no longer a good strategy.